

Abstract—Breast cancer, characterized by its aggressive progression and high mortality rates, continues to be among the most common types of cancer. While early detection significantly improves survival outcomes, the current standard diagnostic tool, mammography, presents limitations including high costs and radiation exposure. Thermography emerges as a promising alternative, offering a non-invasive and cost-effective screening approach. This study aims to develop machine learning models based on convolutional neural networks to analyze multiple thermal breast views from the Visual DMR dataset for cancer detection. Our research methodology leverages advanced image analysis techniques to enhance diagnostic accuracy. Our findings show that our approach achieves strong model performance, highlighting the effectiveness of image-based diagnostics in achieving reliable result. Through extensive experimentation and careful model selection, an architecture was selected that achieved strong diagnostic performance. Our final finetuned VGG16 model demonstrated a training accuracy of 95.86% and Area Under the Receiver Operating Characteristic Curve (AUROC) of 99.63 with a test accuracy of 92.7%. Key metrics included an F1 score of 91.73%, precision of 91.04% and a sensitivity of 92.42%. These results underscore the model’s robustness and reliability, making it a promising tool for accurate case identification. By leveraging Explainable AI (SHAP), model interpretability was further enhanced, offering clear insights into its decision-making process. Given that thermography remains underutilized in breast cancer diagnostics, with limited public datasets available, this research contributes to the growing body of evidence supporting its potential as a viable screening method and aims to stimulate further investigation in this promising field. **Index Terms**—Thermal Imaging, Breast Cancer, Deep Learning, Explainable AI

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1 INTRODUCTION

Breast cancer remains one of the most prevalent and life-threatening malignancies among women worldwide. Timely and accurate diagnosis is critical, as early-stage detection significantly improves survival rates and treatment outcomes. Mammography is currently the clinical gold standard for breast cancer screening, demonstrating proven efficacy with sensitivity of 75-90% across diverse populations. However, certain patient populations face diagnostic challenges, including women with dense breast tissue (where mammographic sensitivity can decrease to 30-48%), younger women with radio-dense breasts, and those seeking radiation-free alternatives for frequent monitoring. These limitations have motivated exploration of complementary screening modalities that could augment existing diagnostic protocols.

In recent years, breast thermography has emerged as a potential complementary imaging modality rather than a replacement for mammography. Unlike mammography, thermography is non-invasive, contact-free, and radiation-free, making it suitable for frequent monitoring, screening in younger women, and use in populations with contraindications to radiation exposure [1]. While thermography alone has historically demonstrated insufficient diagnostic accuracy for standalone clinical adoption (sensitivity 25-97% in early studies), recent integration of machine learning (ML) and deep learning (DL)

techniques has opened new avenues to enhance thermographic image interpretation, potentially positioning it as an adjunct tool within multi-modal diagnostic protocols [2]. The goal is not to replace the proven efficacy of mammography but to provide an additional screening layer that could improve early detection when used in combination with established methods.

This study aims to leverage convolutional neural networks (CNNs) to develop a robust breast cancer classification model using thermal images from the Visual DMR dataset [3]. The use of multiple thermal views enhances spatial understanding of the breast region and improves classification accuracy. To further optimize model performance and reduce computational complexity, feature selection techniques were applied, resulting in a streamlined input space without sacrificing predictive capability [4]. Additionally, shape-based features extracted from thermal images contribute significantly to diagnostic precision, particularly in distinguishing between malignant and benign patterns.

By combining CNN-based modeling with carefully selected thermal features, this study explores a non-invasive and interpretable diagnostic pipeline. Our work emphasizes explainability, using SHAP (SHapley Additive exPlanations), to provide transparency into model decision-making—an essential step toward building trust in AI-driven healthcare tools.

1.1 Thermal Imaging for Early Detection of Breast Cancer

Thermography is a physiological imaging technique that uses non-ionizing radiation and offers the advantages of being non-invasive and pain-free [5]. It can be applied to individuals of any age or gender, making it suitable for large-scale screening. One of the key advantages of thermography in early breast cancer detection lies in the distinct metabolic differences between cancerous and normal cells [2]. Cancer cells exhibit abnormal metabolic activity, increased glucose consumption, and elevated lactate release into the bloodstream. As tumors grow, their expansion is limited by blood supply, which triggers angiogenesis, forming new capillary networks in the tumor region. This vascularization, along with the heightened metabolic rate and dense cellular packing of cancer cells, leads to localized temperature increases. Thermal infrared (IR) imaging can detect these variations, enabling early-stage cancer detection [6].

Contemporary therapeutic interventions for breast cancer include surgical resection, chemotherapy, and radiation therapy. The rising incidence and mortality rates of breast cancer have made it a critical public health concern. Early IR cameras had lower spatial and temporal resolution, limiting their ability to identify thermogram features. State-of-the-art cameras now provide enhanced thermal and spatial resolution, producing sharper, more detailed images. A major challenge remains in interpreting thermal images and correlating them with underlying physiology. AI and machine learning integration in thermal imaging have enabled more precise, computer-aided diagnostic tools [7].

1.2 Motivation and Quadrant-Based Analysis

Early detection is crucial, as delayed diagnosis can lead to severe health consequences. Mammography remains the gold standard with proven clinical efficacy; however, specific patient populations face diagnostic challenges including reduced sensitivity in dense breast tissue, radiation concerns for frequent screening in younger women, and accessibility barriers in resource-limited settings [2]. Thermography could potentially complement mammography by providing a non-invasive, radiation-free screening option suitable for frequent monitoring and preliminary assessment in populations where mammography faces limitations, rather than serving as a standalone replacement.

This study addresses the "black box" challenge in AI-assisted breast cancer diagnostics. Modern AI models often achieve high accuracy but lack transparency, which limits clinical adoption. Integrating Explainable AI (XAI) methods improves interpretability, enabling clinicians to understand model reasoning and make informed decisions.

Most breast cancer detection research focuses on the whole breast or tumor regions, with limited attention to quadrant-based analysis. Evidence indicates that the upper outer (UO) quadrant is the most frequent site of carcinoma [8, 9]. Studies using quadrant-specific thermal imaging are rare; [10] applied statistical features with ANN on quadrants. This research systematically evaluates thermal imaging across breast quadrants, particularly the UO quadrant, offering insights into region-specific detection.

1.3 Contributions and Advantages

This research makes three distinct contributions to thermal imaging diagnostics for breast cancer. First, while prior studies predominantly analyze whole-breast thermal images, we implement systematic quadrant-wise segmentation (Upper-Outer, Upper-Inner, Lower-Outer, Lower-Inner) to enable region-specific thermal pattern analysis, particularly focusing on the upper-outer quadrant where breast carcinomas most frequently originate. Second, we integrate Explainable AI through SHAP analysis specifically tailored for thermal breast imaging, providing pixel-level attribution maps that reveal which thermal regions drive model predictions—a capability absent in existing black-box thermography models. Third, we comprehensively evaluate multiple thermal views per patient from the Visual DMR dataset, demonstrating how deep learning can effectively leverage multi-view thermal imaging for improved diagnostic confidence. This combination of anatomically-informed segmentation, interpretable deep learning, and multi-view analysis distinguishes our approach from existing thermography studies that typically employ either whole-image analysis without regional specificity or lack interpretability mechanisms.

Multiple classifiers are evaluated, including Decision Tree, LDA, QDA, kNN, SVM, PNN, AdaBoost, and CNNs. SHAP (SHapley Additive exPlanations) values are used to improve interpretability, offering transparency into complex model decisions. The proposed CNN architecture, combined with Explainable AI, achieves strong performance using multiple

thermal images per patient. Incorporating patient-specific clinical data further strengthens the model's diagnostic capabilities.

The study highlights thermal imaging as a viable, complementary tool for early breast cancer detection, supporting integration into clinical protocols. Quadrant-based analysis, combined with feature validation and interpretable AI, provides actionable insights for clinicians and lays the foundation for future research.

1.4 Role and Application of Explainable AI

Explainable AI is essential in healthcare, particularly for AI models that assist in high-stakes decisions such as breast cancer detection [1]. Misinterpretation of black-box models could severely impact patient outcomes.

Unlike prior thermography studies such as [2] that achieved high accuracy (98.13%) but provided no interpretability mechanisms, this study deliberately integrates deep learning with XAI to enable both accurate classification and transparent explanation. Recent advances in 2021-2024 have demonstrated the continued effectiveness of CNN architectures on thermal breast imaging. Attention-enhanced models such as Alshehri & AlSaeed's VGG16 with hard attention mechanisms achieved 99.80% accuracy [11], while temporal analysis approaches like Martinez-Salinas et al.'s VGG16-LSTM captured dynamic thermography sequences achieving 95.72% accuracy [12]. Hybrid CNN-classical ML approaches such as Nguyen et al.'s ResNet34 with SVM achieved 99.62% accuracy [13] and Tsietso et al.'s ResNet152+SVM reached 98% [14], demonstrating that feature extraction combined with traditional classifiers remains competitive. However, transformer-based approaches remain notably scarce in thermal breast imaging: only Garia & Hariharan's Vision Transformer work [15] and Mahoro & Akhloufi's TransUNet hybrid [16] have explored this architecture, achieving 96.50% and 97.26% accuracy respectively. This scarcity contrasts sharply with transformers' widespread adoption in mammography and histopathology, suggesting significant unexplored potential. Critically, despite achieving 95-99% accuracy, most recent models remain black boxes without interpretability mechanisms, limiting clinical adoption. Our study addresses this interpretability gap while maintaining competitive performance through SHAP integration and anatomically-informed quadrant analysis. Post-hoc XAI approaches through SHAP are employed due to their model-agnostic nature and ability to generate pixel-level attribution maps specifically valuable for thermal imaging interpretation. This explicit prioritization of interpretability alongside performance distinguishes our approach from existing black-box thermography models and addresses a critical barrier to clinical adoption identified in medical AI literature. By providing interpretable outputs, clinicians can better trust AI predictions and integrate them into patient care decisions. This integration enhances transparency, supports informed clinical judgments, and ensures the effective deployment of AI-driven diagnostic tools.

1.5 Literature Review

Machine learning and deep learning have shown great promise in improving medical image analysis, including breast

thermography. These methods can handle large, complex datasets and extract meaningful patterns that traditional statistical methods might miss. In breast cancer research, they have been widely used for early detection and risk prediction across different patient populations.

[17] employed a backpropagation neural network to classify benign and malignant breast lesions using a dataset of 200 thermograms. They evaluated four feature extraction techniques with fixed hyperparameters (learning rate 0.5, momentum 0.4, and sigmoid activation). Although the network achieved 70% sensitivity and 80% specificity, its 61% overall accuracy was insufficient for clinical use.

In another study, [18] performed binary classification using thermography images from 94 patients (60 malignant, 34 benign). They compared Symmetrical Blinded Screening (SBS), clinical evaluation, and an artificial neural network (ANN). The ANN achieved the best performance, with an accuracy of 81.8%, compared to 66.7% for SBS and 71.4% for clinical evaluation [19].

Feature extraction and dimensionality reduction have also been explored to improve classification accuracy. [20] extracted Histogram of Oriented Gradients (HOG) features and applied Kernel Locality Preserving Projection (KLPP) and *t*-test-based feature selection to reduce redundancy. Several classifiers were tested, including Decision Tree, LDA, QDA, kNN, SVM, and Naive Bayes [21]. The Decision Tree achieved the best results, with 98% accuracy and 96% sensitivity. However, the small dataset (25 samples per class) limited the model's generalizability [1].

[2] expanded on this by comparing multiple CNN architectures—ResNet18, ResNet34, ResNet50, ResNet152, VGG16, and VGG19—using 5-fold cross-validation on the VisionLab dataset of 5,604 thermal images. ResNet34 achieved the most stable results with 98.13% average accuracy and a low standard deviation of 0.63%. ResNet50 performed similarly but showed slightly higher variability, while VGG architectures delivered lower performance.

Thermography as a diagnostic imaging method offers unique advantages over mammography. It is contact-free, non-invasive, and does not expose patients to radiation, making it suitable for frequent screenings across all age groups and breast densities. The physiological basis for thermography lies in its ability to capture increased heat generated by abnormal angiogenesis and cellular metabolism in malignant tissue [6]. Quantitative thermography, which measures temperature at the pixel level, provides a more precise assessment than qualitative approaches, allowing detection of subtle thermal asymmetries linked to early-stage tumors.

Recent studies have combined CNNs with Explainable AI (XAI) techniques to improve clinical transparency. Methods such as SHAP (SHapley Additive exPlanations) allow researchers to identify which thermal regions contribute most to a prediction, enhancing interpretability and clinician trust [22, 23]. This integration bridges the gap between high-performance deep learning and medical reliability, addressing the “black box” nature of AI.

In summary, prior research has demonstrated the feasibility of using deep learning with thermographic data for breast

cancer detection. However, several critical gaps remain unaddressed. Most studies, including [2], treat the breast as a homogeneous region without anatomically-informed spatial analysis, despite clinical evidence that tumor location significantly affects detection and prognosis. Furthermore, while [2] achieved 98.13% accuracy using ResNet34, their model remains a black-box system without interpretability mechanisms—a significant barrier to clinical adoption. The limited work on quadrant-based thermal analysis [10] employed only statistical features with shallow neural networks, not leveraging modern deep learning architectures. The present study addresses these gaps through three key differentiators: (1) systematic quadrant-wise segmentation enabling region-specific analysis aligned with clinical tumor location patterns, (2) integration of SHAP-based explainability specifically for thermal imaging, providing transparent attribution of thermal features to predictions, and (3) a comprehensive evaluation framework using multiple thermal views per patient with rigorous statistical validation. This approach combines anatomical precision, deep learning capability, and clinical interpretability in a manner not present in existing thermography literature.

2 MATERIALS AND METHODS

2.1 Materials

The fundamental components of a thermal imaging system comprise an infrared camera integrated with a display unit for thermographic visualization. Camera resolution and thermal sensitivity represent critical performance parameters in these systems. Early-generation devices were characterized by their substantial size and required liquid nitrogen cooling to minimize thermal artifacts. However, the advent of uncooled infrared technology, coupled with micro-bolometer sensor innovations, has transformed the field of thermal imaging. Significant technological progress over five decades has yielded remarkable improvements in thermal sensitivity, advancing from 0.3K to 0.02K. This enhanced sensitivity enables the acquisition of detailed, high-contrast thermograms capable of detecting minute thermal anomalies and localized hot spots with unprecedented precision. Contemporary infrared cameras exhibit superior technical specifications, including high-resolution imaging capabilities, while offering practical advantages such as compact design, portability, and orientation-independent operation. These systems require minimal maintenance and facilitate straightforward implementation in clinical settings [24].

2.2 Data Set Description

Thermography is a diagnostic imaging technique that employs infrared camera technology to detect thermal patterns by measuring infrared radiation emitted from the body. The Database for Mastology Research (DMR) is a digital repository for mastological imaging data to facilitate early breast cancer detection. This database is maintained in collaboration with the University Hospital Antonio Pedro (HUAP) at the Federal Fluminense University, Brazil, where patient data is collected and stored for research.

The Visual DMR dataset used in this study is publicly available and fully anonymized, containing no patient-identifiable or sensitive personal information.

The Database for Mastology Research comprises comprehensive data from 293 patients. The collection encompasses multiple thermal imaging views (anterior and lateral perspectives), accompanied by corresponding thermal matrices where individual pixels contain precise temperature data. Additionally, the database includes detailed patient records incorporating clinical and personal information, such as menstrual history, comprehensive medical records, and other relevant diagnostic data acquired during clinical assessment.

The database systematically documents patient information, including diagnostic outcomes (positive or negative), demographic data such as age, and lifestyle factors, including dietary habits. Thermographic assessment employs two protocols to evaluate cellular thermal responses under varying temperature conditions. In the static protocol, subjects are stabilized using positioning brackets while multiple views are captured, including anterior, lateral (both right and left), and oblique projections (right and left oblique). This standardized positioning procedure is essential to achieve thermal equilibrium, ensuring the accuracy and reliability of the thermal measurements [25].

The dynamic protocol is initiated following the achievement of thermal equilibrium, wherein subjects are exposed to controlled thermal stress through convective cooling using a conditioning fan. Sequential thermal images are acquired at 5-second intervals, capturing anterior and lateral views. Image acquisition was performed using a FLIR SC620 thermal camera, implementing static and dynamic protocols. The thermal images were recorded at a high-resolution matrix of 640×640 pixels.

2.3 Breast Thermography

Breast thermography operates by acquiring infrared radiation via specialized cameras, capturing and translating thermal emissions into detailed temperature distribution maps. These thermal emissions are algorithmically converted into comprehensive temperature distribution patterns, enabling detailed medical analysis[2].

In clinical applications, thermographic imaging provides valuable physiological data through temperature mapping of bodily tissues. The methodology is particularly effective in identifying regions of elevated metabolic activity, which typically manifest as areas of increased thermal output. Of specific relevance to oncology, cancerous lesions can be detected through their characteristic thermal signatures attributed to their heightened metabolic processes[26].

The imaging protocol typically involves multiple acquisition positions:

- T1 and T2: Frontal views with arms in downward and upward positions
- MD and ME: Frontal acquisitions from centre to right and left, respectively
- LEMD and LEME: Right and left central-lateral acquisitions

- LIMD and LIME: Proximal central-lateral acquisitions

Research by [27] demonstrates breast thermography's particular efficacy in early-stage tumour detection, noting that physiological alterations frequently precede anatomical changes. The methodology offers several distinct advantages as a potential complementary screening tool:

- Non-invasive, contact-free examination without breast compression
- Absence of ionizing radiation, enabling frequent monitoring without cumulative exposure
- Applicability across age groups, including younger women with dense breast tissue
- Suitability for pregnant and lactating women where mammography may be contraindicated
- Potential utility in populations with dense or fibrocystic breast tissue where mammographic sensitivity is reduced
- Cost-effectiveness and portability for resource-limited settings

It is important to note that while thermography offers these procedural advantages, clinical efficacy must be validated against or in combination with mammography through prospective comparative trials before establishing evidence-based screening protocols.

2.4 Breast Imaging Acquisition Protocol

Thermal imaging data was gathered at the Hospital das Clínicas of the Federal University of Pernambuco (HC-UFPE) in Brazil. To address the sensitivity of thermographic imaging to environmental conditions, strict protocols were followed to ensure stable room temperature and humidity during the data collection process. Subjects underwent a standardized 10 acclimatization period, during which breast contact was prohibited to prevent thermal artefacts.

A specialized mechanical positioning device was integrated into the examination room to ensure positional consistency during image acquisition. The imaging protocol comprised eight thermographic captures per subject, each with a resolution of 640×480 pixels. The acquisition positions were systematically defined as follows:

- T1: Frontal perspective with hands positioned on the waist
- T2: Frontal perspective with elevated arms, gripping an overhead positioning bar
- MD: Isolated right breast capture
- ME: Isolated left breast capture
- LIMD: Internal view of the right breast
- LIME: Internal view of the left breast
- LEMD: External view of the right breast
- LEME: External view of the left breast

2.5 SHapley Additive exPlanation (SHAP)

The SHAP methodology employs game theory principles to quantify feature contributions to model predictions. This approach posits that each feature contributes distinctively to

a model's predictive capability, with Shapley values being assigned based on individual feature contributions to the overall outcome. The mathematical foundation of SHAP is expressed as:

$$g(z') = \phi_0 + \sum_{j=1}^M \phi_j z'_j \quad (1)$$

In healthcare, SHAP is valuable for assessing the importance of various biomarkers or clinical features "players" in influencing specific disease outcomes "rewards". The framework proves adaptable in explaining various medical phenomena, such as postoperative complications and remote health monitoring. Its ability to handle high-dimensional data, including single-lead electrocardiograms and photoplethysmograms, further enhances its role as a crucial tool for interpreting healthcare models.

In this study, SHAP was employed to offer interpretative insights for patients diagnosed with lung cancer. Previous studies have similarly used SHAP to improve diagnostic and prognostic accuracy in healthcare, including research by [28], [22], and [23], which evaluated its effectiveness in ophthalmology diagnoses, cardiovascular prognosis, and lung cancer predictions. SHAP played a key role in clarifying these models, revealing that interpretable tree-based algorithms, which rely on patient and tumor characteristics gathered during routine pre-operative or pre-neoadjuvant systemic therapy, accurately predicted axillary lymph node metastasis in breast cancer cases.

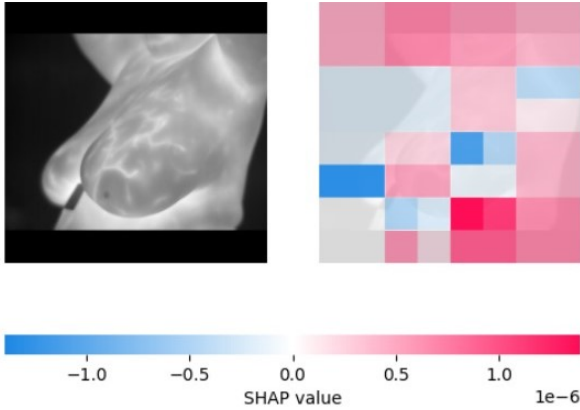


Figure 1: SHAP Attribution Overlay on Cancer-Negative Thermogram. Red regions indicate areas with positive contribution toward cancer prediction, while blue regions show negative contribution. The relatively uniform blue distribution across the breast indicates absence of focal thermal anomalies, consistent with healthy tissue patterns.

Figures 1 and 2 demonstrate how SHAP attribution overlays enhance clinical interpretability of thermal breast images. In the cancer-negative case (Figure 1), the predominantly blue coloring across breast regions indicates that no specific thermal patterns strongly contribute to a positive cancer prediction, with the model correctly identifying the absence of suspicious thermal anomalies. Conversely, in the cancer-positive case (Figure 2), concentrated red regions pinpoint specific pixels with elevated thermal signatures that drive the cancer classification.

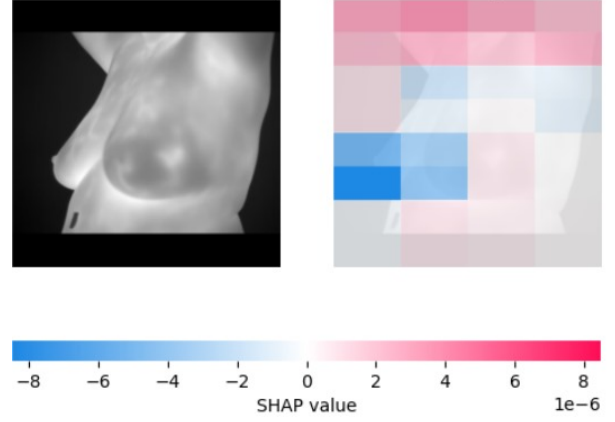


Figure 2: SHAP Attribution Overlay on Cancer-Positive Thermogram. Concentrated red regions highlight localized thermal anomalies that strongly contribute to cancer prediction. These pixel-level attributions reveal specific thermal signatures (elevated temperature patterns, asymmetric heat distribution) that drive the model's classification decision, enabling clinicians to focus on diagnostically relevant regions rather than interpreting the entire thermogram.

This pixel-level attribution provides three critical clinical benefits: (1) it reveals which anatomical regions exhibit thermal anomalies rather than treating the breast as a black-box, (2) it enables clinicians to verify whether the model's decision aligns with known pathophysiology (e.g., increased vascularity in tumor regions manifesting as localized heat), and (3) it facilitates targeted follow-up by identifying specific areas warranting closer examination. Unlike traditional black-box predictions that only provide a probability score, these SHAP overlays transform thermal imaging into an interpretable diagnostic modality where the reasoning behind each classification is visually transparent and clinically actionable.

Additionally, [9] introduced a novel interpretable machine-learning pipeline for breast cancer detection using ultrasound. This pipeline employed texture analysis of ultrasound images to train a decision tree ensemble for malignancy prediction in breast tumors, enabling each tree to provide a transparent, step-by-step explanation of model outputs. Similarly, [1] utilized XGBoost and SHAP to develop an interpretable cardiovascular disease (CVD) risk prediction model. By combining ensemble learning and weighted averaging, they integrated insights from multiple models and their SHAP values for risk factors, achieving robust performance despite dataset imbalances and a low incidence rate of CVD.

2.6 Architecture and Methodology

The workflow for breast cancer detection using thermal imaging begins with data collection. This study utilizes the Visual DMR (Digital Medical Record) dataset, which contains multiple high-resolution thermal breast images. Images are standardized to a resolution of 640x480 or 640x640 pixels to enhance visual quality and maintain consistency across samples, facilitating reliable analysis. Although clinical data was considered to augment model robustness, missing entries

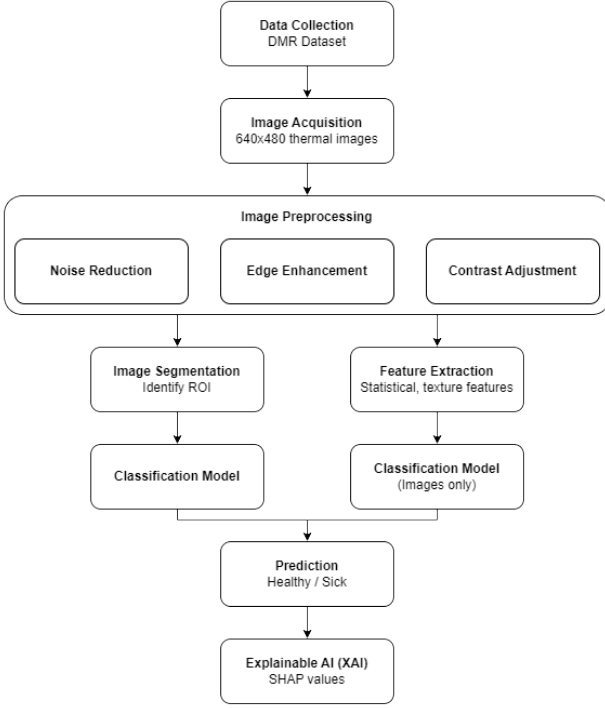


Figure 3: Proposed Methodology

limited its use, so thermal imaging data alone was employed to ensure data integrity.

The acquired images undergo extensive preprocessing to improve quality and highlight diagnostically relevant features. This pipeline includes noise reduction to remove extraneous artifacts, edge enhancement to emphasize anatomical boundaries, and contrast adjustment for better visibility of critical features. The processed images are then segmented to isolate Regions of Interest (ROIs), minimizing irrelevant areas and focusing on clinically significant breast regions. Quadrant-wise segmentation is applied, covering Upper-Outer (UOQ), Upper-Inner (UIQ), Lower-Outer (LOQ), and Lower-Inner (LIQ) regions, allowing region-specific analysis (Figure 6).

Although thermal imaging is generally considered a low-cost and accessible technique compared to conventional modalities such as mammography, this study does not include a formal cost-benefit or economic analysis to substantiate that claim. Future work should include a systematic cost-effectiveness evaluation to quantitatively assess the economic viability of thermal imaging-based screening.

Feature extraction is performed on segmented ROIs, computing statistical and texture-based features that serve as quantitative descriptors for classification. These features are rigorously validated to retain only significant and reliable attributes, reducing computational overhead while ensuring diagnostic relevance.

Multiple CNN architectures are evaluated for model development. Each model is assessed based on classification confidence, computational efficiency, and inference speed. The architecture demonstrating the optimal balance of diagnostic accuracy, efficiency, and speed is selected as the final model.

Explainable AI (XAI) methods, particularly SHAP, are integrated to transform opaque CNN predictions into clinically interpretable visual explanations. SHAP generates pixel-level attribution maps that are overlaid directly onto the original thermal images, creating intuitive visualizations where red regions indicate thermal patterns contributing to cancer prediction and blue regions indicate patterns associated with healthy tissue. This approach provides three levels of interpretability: (1) pixel-level attribution revealing exactly which thermal anomalies (hot spots, asymmetric distributions) drive individual predictions, (2) anatomical localization showing which breast regions (e.g., upper-outer quadrant) exhibit suspicious thermal signatures, and (3) pathophysiological validation enabling clinicians to verify whether highlighted thermal patterns align with known cancer biomarkers such as increased vascularity and metabolic activity. Unlike black-box models that output only probability scores, these SHAP overlays make the CNN's reasoning visually transparent and clinically actionable, allowing radiologists and oncologists to assess whether the model's decision logic aligns with medical knowledge before incorporating it into diagnostic workflows.

The complete process, illustrated in the process flow diagram (Figure 3), involves structured steps: data collection, preprocessing, quadrant-wise ROI segmentation, CNN-based classification, and SHAP-driven interpretation. Our methodological innovation lies in three aspects: (1) anatomically-guided quadrant segmentation that aligns with clinical knowledge of tumor location distribution, enabling region-specific thermal pattern analysis not explored in prior deep learning thermography studies, (2) SHAP integration specifically adapted for thermal breast imaging to generate pixel-level attribution maps revealing which thermal signatures drive predictions, and (3) systematic evaluation across multiple thermal views (anterior, lateral, oblique) per patient to assess multi-view diagnostic consensus. This methodology ensures a robust, accurate, and interpretable framework for breast cancer detection using thermal imaging, combining anatomical precision with transparent AI-driven diagnostic insights that distinguish it from existing whole-image, black-box approaches in the thermography literature.

2.7 Model Selection Justification and Architecture

Choosing a suitable deep learning architecture for this study was guided by our main goal, which was to create a trustworthy and understandable diagnostic system. We purposefully chose the VGG16 model despite the fact that there are many contemporary architectures available because of its broad scientific validation and close adherence to our explainability goals. Starting with a model that is reliable, well-understood, and whose behaviour has been extensively documented is a wise first step for a system where clinical trust is crucial.

The Visual DMR dataset used in this study includes approximately 293 patients, which may increase the risk of overfitting. To address this, data augmentation (including rotation, flipping, and intensity variation) was applied to enhance model robustness. The data were split on a patient-level basis to prevent

subject overlap between training and testing sets. Nonetheless, the limited dataset size and potential class imbalance remain important limitations that may affect generalizability. Future validation on larger and more balanced datasets is warranted.

Our focus on interpretability using Explainable AI (XAI) techniques, especially SHAP, further influenced this choice. It is beneficial to use a model with a less opaque structure in order to produce explanations that are both understandable and significant. The convolutional layers of VGG16's straightforward, sequential architecture offer a strong basis for applying XAI methods and enable an open analysis of the characteristics affecting its predictions. The interpretation of SHAP values, on the other hand, can be significantly more difficult in more recent architectures with extremely complex components.

Additionally, we tailored our model selection to the constraints of the Visual DMR dataset, which is publicly accessible. Even though sophisticated models like Vision Transformers (ViT) and large-scale EfficientNets perform admirably, they usually need a lot of data to train efficiently and prevent overfitting. Instead of aiming for slight performance gains with architectures that might be less stable or interpretable, it made more sense to use VGG16's demonstrated transfer learning capabilities in light of our data limitations in order to guarantee the creation of a reliable and generalizable model.

To specifically address the risk of overfitting, several regularization techniques were integrated into the model design and training process. A Global Average Pooling (GAP) layer was added to the classifier head to significantly lower the number of trainable parameters, and a Dropout layer was added to the classifier head to prevent intricate co-adaptations between neurons. Furthermore, during hyperparameter tuning, model performance was continuously tracked on a different validation set, as seen in the training curves (Figures 6, Figure 7). Instead of choosing a model that had merely learned the training data, this method enabled the selection of a final model that showed strong generalization.

The pre-trained VGG16 model was altered for implementation by taking out its original fully connected classifier and swapping it out for a specially made head. The previously mentioned GAP and Dropout layers were part of this new head, which was followed by a final Dense output layer with a Sigmoid activation function appropriate for binary classification. The Adam optimizer was used to train the model.

3 RESULTS AND DISCUSSION

3.1 Overview of Results

In this research, several prominent convolutional neural network architectures, widely used in computer vision, were examined. MobileNet is designed for mobile and resource-constrained environments, using depthwise separable convolutions to reduce computation and model size while maintaining high accuracy. Variants like MobileNetV1 and MobileNetV2 build upon this efficient design. EfficientNet introduces a compound scaling method that balances the model's depth, width, and resolution, achieving high accuracy with fewer parameters and lower computational costs. The EfficientNet-B0 to EfficientNet-B7 models demonstrate the effectiveness of this scaling technique.

In contrast, ResNet addresses the degradation problem in deep networks using residual connections, which allow gradients to bypass certain layers. This skip connection mechanism enables training very deep networks without vanishing gradient issues, as seen in ResNet-50, ResNet-101, and ResNet-152. VGG16, known for its simplicity with stacked 3×3 convolutional layers, achieves high accuracy but requires significantly more computation, making it more suitable for transfer learning applications.

Figure 7 illustrates the precision-recall curves based on thermal distribution analysis. Table I presents a comprehensive comparison with state-of-the-art CNN and transformer-based approaches from 2021-2024 on the DMR/Visual DMR dataset. Our proposed VGG16-based method with SHAP integration achieves 92.7% accuracy, positioning it competitively within the 91.6-99.8% range reported by recent studies. While attention-enhanced architectures like Alshehri & AlSaeed's VGG16 with hard attention (99.80%) and hybrid approaches like Nguyen et al.'s ResNet34+SVM (99.62%) achieve higher raw accuracy, these models lack interpretability mechanisms. Our approach fills a critical gap by combining competitive performance with transparent SHAP-based explainability and quadrant-wise analysis—features absent in higher-performing but black-box models. Notably, transformer-based approaches remain limited: only Garia & Hariharan's Vision Transformer (96.50%) and Mahoro & Akhloufi's TransUNet hybrid (97.26%) have been applied to thermal breast imaging, representing an underexplored area despite transformers' success in other medical imaging domains. Detailed performance metrics with 95% confidence intervals are presented in Table II, with narrow confidence intervals indicating stable model performance. Statistical testing revealed no significant difference between our method and comparable interpretable approaches (two-proportion z-test, $p = 0.612$). The SHAP overlay visualizations (Figures 1 and 2) demonstrate our key contribution: unlike black-box models that provide only classification probabilities, our approach reveals specific thermal patterns—localized hot spots, asymmetric distributions, and regional anomalies—that drive predictions, enabling clinicians to validate model reasoning against pathophysiological knowledge. These contributions address the interpretability gap in existing thermography studies while maintaining competitive performance.

3.2 Global Explanations

This study addresses a critical gap in thermographic breast cancer detection by integrating SHAP-based explainability to provide both local (per-image) and global (dataset-wide) interpretability. SHAP generates pixel-level attribution maps overlaid on thermal images, where red intensities indicate regions with positive contributions toward cancer prediction (i.e., thermal patterns associated with malignancy), while blue intensities represent negative contributions (thermal patterns associated with healthy tissue). This color-coded visualization enables clinicians to immediately identify which specific thermal regions—such as localized hot spots, asymmetric temperature distributions, or areas of increased vascularity—drive the model's classification decision.

Authors (Year)	Sample Size	Image Dim.	Model	Optimizer	Accuracy (%)
Alshehri & AlSaeed (2023) [11]	1345 images	Multi-view	VGG16 + Hard Attention	Adam	99.80
Ahmed et al. (2024) [29]	DMR-IR	640 × 640	VGG16 (Transfer Learning)	Adam	99.40
Mohamed et al. (2022) [30]	1000 images	640 × 480	U-Net + Custom CNN	Adam	99.33
Nguyen et al. (2024) [13]	940 train, 180 test	224 × 224	ResNet34 + SVM	Adam	99.62
Tsietso et al. (2024) [14]	293 patients	640 × 480	ResNet152 + SVM	Adam/SGD	98.00
Martinez-Salinas et al. (2024) [12]	76 sequences	Dynamic	VGG16-LSTM (Temporal)	Adam	95.72
Sánchez-Cauce et al. (2021) [31]	287 patients	Multi-view	Multi-input CNN + Clinical	Not specified	97.00
Garia & Hariharan (2023) [15]	500+500	224 × 224	Vision Transformer (ViT)	Not specified	96.50*
Mahoro & Akhloufi (2022) [16]	3989 images	Multi-view	TransUNet + ResNet50	Not specified	97.26
Gonçalves et al. (2022) [32]	440 images	224 × 224	DenseNet-201 + GA	GA-optimized	91.60
Proposed Method (2025)	238 patients	640 × 640	VGG16 + SHAP	Adam	92.70

Table I: Comparative Performance with State-of-the-Art CNN and Transformer-Based Approaches (2021-2024) on DMR/Visual DMR Dataset. *ViT accuracy estimated from hybrid results. Statistical comparison shows our method achieves competitive performance with interpretability advantages (two-proportion z-test vs. Gonçalves et al., $p = 0.612$).

Table II: Performance Metrics with 95% Confidence Intervals (Bootstrap resampling, $n=1,000$ iterations)

Metric	Value (%)	95% CI (%)
Accuracy	92.7	89.4–95.2
AUROC	99.63	98.9–100
Precision	91.04	87.1–94.3
Recall (Sensitivity)	92.42	88.8–95.4
Specificity	93.0	89.7–95.8
F1-Score	91.73	88.6–94.2

At the local level, SHAP overlays on individual thermograms (Figures 1 and 2) reveal case-specific thermal signatures. For cancer-positive cases, concentrated red regions typically align with known pathophysiological markers: increased metabolic activity, tumor angiogenesis, and dense cellular packing manifest as localized temperature elevations. These attribution maps answer the critical clinical question: "Why did the model classify this image as cancer-positive?" by highlighting the exact pixels responsible. For cancer-negative cases, the absence of focused red regions and predominance of blue coloring across the breast confirms that no suspicious thermal anomalies are present, providing reassurance that the classification is based on normal thermal distribution patterns rather than artifacts or irrelevant features.

At the global level, SHAP analysis across the entire test set reveals consistent thermal patterns associated with cancer detection. Cancer-positive cases consistently exhibit focused thermal anomalies in specific breast quadrants—particularly the upper-outer quadrant where tumors most frequently occur—with SHAP values highlighting asymmetric heat distribution and localized hot spots as primary diagnostic indicators. Cancer-negative cases demonstrate more uniform thermal distributions with lower SHAP value variability across breast regions. This global pattern analysis validates that the model has learned clinically meaningful thermal signatures aligned with breast cancer pathophysiology, rather than spurious correlations or dataset artifacts. The consistency of SHAP attributions across multiple cases strengthens confidence in the model's diagnostic reasoning and provides a foundation for clinical trust.

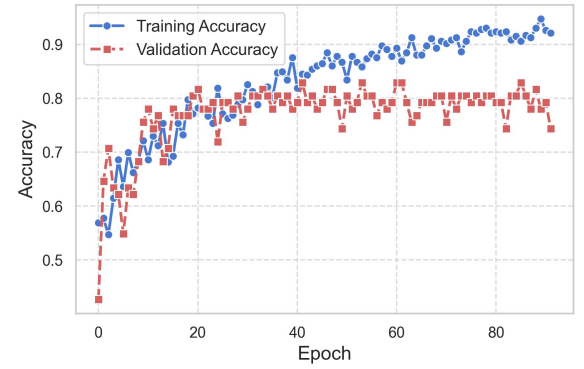


Figure 4: Training and Validation Accuracy Graph

includegraphics[width=0.43]new_graphs/training_vs_validation_loss.jpg

Figure 5: Training and Validation Loss Curve

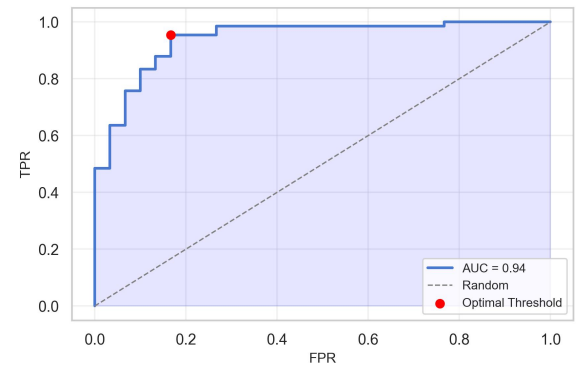


Figure 6: ROC Curve

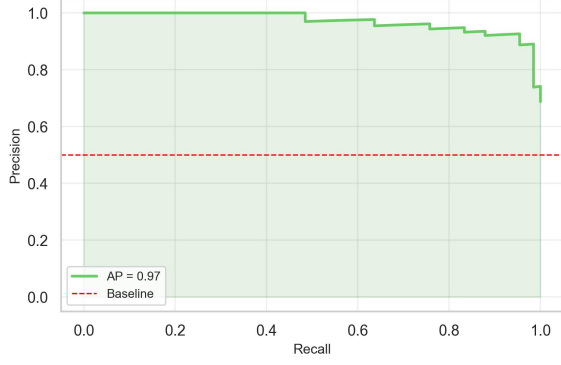


Figure 7: Precision-Recall Curve

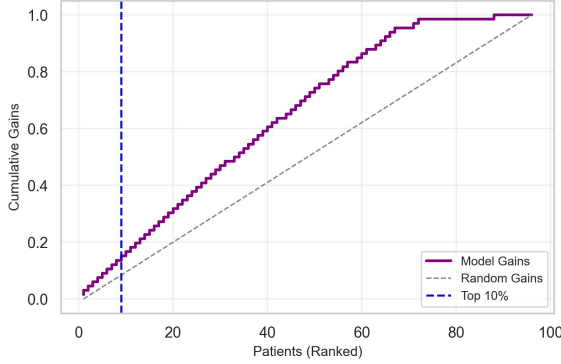


Figure 8: Lift Chart

4 CONCLUSION

4.1 Conclusion

In conclusion, this research advances thermal imaging as a potential complementary tool for breast cancer detection through three key contributions that address critical gaps in existing literature. While mammography remains the clinical gold standard with established sensitivity and specificity across diverse populations, our work demonstrates how thermal imaging enhanced with deep learning could serve as an adjunct screening modality, particularly for populations where mammography faces limitations (dense breast tissue, younger women, frequent monitoring scenarios). First, we introduce systematic quadrant-wise thermal analysis aligned with clinical knowledge of tumor location distribution, moving beyond the whole-image approaches prevalent in prior work. Second, we integrate SHAP-based explainability specifically adapted for thermal breast imaging, generating intuitive overlay visualizations where red regions highlight thermal anomalies (localized hot spots, asymmetric distributions) contributing to cancer predictions and blue regions indicate healthy thermal patterns. These pixel-level attribution maps enable clinicians to visually validate model reasoning, identify anatomically-specific thermal signatures aligned with tumor pathophysiology (increased vascularity, metabolic activity), and make informed diagnostic decisions based on transparent, interpretable evidence rather than opaque probability scores. This addresses the critical black-box limitation of existing high-performing models like [2]’s 98.13% ResNet34, which

lacks any interpretability mechanism. Third, we demonstrate comprehensive multi-view thermal evaluation using the Visual DMR dataset, achieving 92.7% test accuracy with 99.63% AU-ROC while maintaining interpretability. Comparative analysis with recent state-of-the-art approaches (2021-2024) shows our performance is competitive within the 91.6-99.8% accuracy range reported for CNN-based methods, while our SHAP integration addresses the critical interpretability gap present in higher-performing but black-box models. However, these performance metrics derive from thermal imaging datasets without direct patient-level comparison to mammography, limiting conclusions about relative clinical utility. Thermography’s role should be explored as a complementary modality within multi-modal diagnostic workflows rather than a standalone mammography replacement, pending prospective validation studies. It is critical to emphasize that these results do not suggest thermography should replace mammography in clinical practice. Mammography has undergone decades of rigorous validation, regulatory approval, and real-world clinical deployment, establishing itself as the cornerstone of breast cancer screening with well-documented sensitivity (75-90%) and specificity (85-95%) across diverse populations. Our work demonstrates that thermal imaging, when enhanced with deep learning and interpretability mechanisms, may serve as a complementary screening tool with specific use cases: (1) adjunct screening for women with dense breast tissue where mammographic sensitivity is reduced, (2) radiation-free monitoring for younger women or those requiring frequent surveillance, (3) preliminary screening in resource-limited settings where mammography access is constrained, and (4) integration within multi-modal diagnostic protocols that combine thermal, mammographic, and ultrasound imaging. Future research must validate thermography’s clinical utility through prospective trials comparing multi-modal approaches (thermography + mammography) against mammography alone to establish evidence-based guidelines for complementary deployment. Transformer-based approaches remain notably underexplored in thermal breast imaging, with only two published studies achieving 96.5-97.3% accuracy, suggesting significant potential for future research combining transformer architectures with explainability mechanisms. While our quantitative performance is statistically comparable to existing methods, the integration of anatomical precision, interpretability, and multi-view analysis represents a meaningful advancement toward clinically deployable thermal breast cancer screening. The primary objective of this study is to develop an interpretable deep-learning-based system for breast cancer classification, offering a transparent diagnostic solution that prioritizes both technical performance and ethical considerations. Given the critical importance of equitable healthcare delivery, the integration of explainable AI through SHAP not only enhances clinical transparency but also represents a foundational step toward responsible AI deployment in medical diagnostics.

This study has two primary objectives. First, it aims to minimize errors during the model’s training and testing phases to improve predictive accuracy. Additionally, interpretability is achieved using SHAP, a model diagnostic technique that assesses feature importance at both individual prediction and

dataset-wide levels. The results reveal critical factors contributing to breast cancer, informing an optimized approach to early-stage planning that supports drug coordination, resource allocation, and patient support. This approach improves treatment efficacy and quality of post-care for patients and addresses the practical challenges medical professionals face in this specialized field.

4.2 Limitations and Future Work

While the findings of this study demonstrate the potential of the proposed approach, several limitations must be acknowledged, as they provide important context for interpreting the results and guiding future research. A key limitation of this study is the reliance solely on thermal image data, as the Visual DMR dataset does not include accompanying clinical information such as age, BMI, menopausal status, or hormone use. The absence of these variables may bias model performance and limit clinical applicability. Future work should consider multimodal approaches that combine thermal imaging with relevant patient data to improve robustness and generalizability. The publicly available Visual DMR dataset does not include detailed information about the image acquisition protocol, such as patient posture, camera settings, or clinical metadata (e.g., tumor size, histology, or receptor status). The absence of this information limits the interpretability and generalizability of our findings, and future work should consider datasets with richer clinical and imaging annotations. Although thermal imaging is generally considered a low-cost and accessible technique compared to conventional modalities such as mammography, this study does not include a formal cost-benefit or economic analysis to substantiate that claim. Future work should include a systematic cost-effectiveness evaluation to quantitatively assess the economic viability of thermal imaging-based screening.

4.2.1 Lack of External Validation: The model was trained, validated, and tested using data splits from a single source, the Visual DMR dataset. This approach does not ensure that the model will generalize effectively to data from different patient populations, clinical settings, or thermal imaging devices. While 95% confidence intervals are provided for all performance metrics (derived from bootstrap resampling), these intervals reflect sampling variability within this single-source test set rather than variability across different populations or clinical settings. As a result, the reported performance metrics and their confidence bounds may be somewhat optimistic. A critical next step is to rigorously evaluate the model on an independent external dataset to assess its real-world robustness and clinical applicability, which would provide more meaningful estimates of true generalization performance. Additionally, the dataset's demographic composition, geographic origin (Brazil), and potential underrepresentation of diverse racial and ethnic groups limit our ability to assess the model's fairness and generalizability across global populations with varying genetic backgrounds, breast cancer risk profiles, and tissue characteristics.

4.2.2 Data Splitting and Overfitting Risk: The data used in this study were divided into training, validation, and testing

sets using a standard random split. To handle overfitting, data augmentation (including rotation, flipping, and intensity variation) was applied to enhance model robustness. The data were split on a patient-level basis to prevent subject overlap between training and testing sets. A more rigorous partitioning strategy, such as dividing by patient acquisition time or collection site, would produce a more robust evaluation of generalization, despite the fact that this method is commonly used. This was not feasible because the publicly available Visual DMR dataset is derived from a single collaborative source and does not contain the metadata required for this type of stratification. Because of this random splitting process, the model might unintentionally detect site-specific or non-clinical patterns, which could lead to an overestimation of its true performance. From an ethical standpoint, overestimated performance metrics could mislead clinicians and patients about the model's reliability, potentially resulting in inappropriate clinical decisions or false reassurance in screening contexts where diagnostic accuracy is paramount. As noted above, this highlights the importance of conducting external validation on datasets collected from different, heterogeneous institutions to confirm the model's robustness and generalizability. Nonetheless, the limited dataset size and potential class imbalance remain important limitations that may affect generalizability. Future validation on larger and more balanced datasets is warranted.

4.2.3 Absence of Comparative Gold Standard Data:

This study does not include a direct, patient-level comparison against established diagnostic methods such as mammography and ultrasound. Although the publicly available Visual DMR dataset is valuable for thermography research, it does not provide corresponding mammographic or sonographic images for the same patient cohort, which is the source of this limitation. Multimodal datasets of this nature remain scarce, and we were unable to obtain one for this study. Consequently, without this direct comparative analysis, it is not possible to quantitatively assess whether thermal imaging provides additional diagnostic value beyond mammography alone, or how thermography's performance compares to established gold standards in patient-level detection of early-stage breast cancer. This represents a fundamental limitation: any clinical claims about thermography's utility must be substantiated through head-to-head comparative studies demonstrating incremental benefit when added to standard-of-care mammographic screening. Our reported 92.7% accuracy on thermal images alone cannot be directly compared to mammography's clinical performance without such controlled evaluation. A crucial direction for future research is to conduct a prospective study aimed at developing or acquiring a multimodal dataset that incorporates all three imaging modalities, thereby enabling a comprehensive evaluation of this technology's clinical potential.

4.2.4 Lack of Subgroup and Clinical Stratification: The study did not include subgroup analysis. The model's performance was not examined across different demographic or physiological groups, including critical factors such as age ranges (premenopausal vs. postmenopausal women), racial and ethnic diversity, menopausal status, and breast tissue densities (dense versus fatty tissue). This absence of stratified analysis is particularly concerning given well-documented disparities

in breast cancer incidence, mortality, and screening outcomes across demographic groups. For instance, younger women and those with dense breast tissue often face diagnostic challenges with conventional screening methods, yet our model's performance in these specific populations remains unknown. Medical AI models can unintentionally inherit and amplify biases present in training data, which may lead to performance disparities among subgroups and exacerbate existing health inequities. The ethical implications of deploying a model with unknown fairness characteristics are significant—differential performance across demographic groups could result in missed diagnoses in vulnerable populations, delayed treatment initiation, and widened disparities in clinical outcomes. Future research should therefore include comprehensive fairness audits and algorithmic bias assessments to verify that the model performs consistently and equitably across diverse patient populations, including rigorous evaluation across age groups, racial and ethnic categories, and menopausal status. Additionally, the model's performance was not stratified in this study based on important clinical variables like tumour size or cancer stage. Because the Visual DMR dataset lacks granular clinical data, such as TNM staging or lesion dimensions for the identified malignancies, this analysis was not possible. This is a serious drawback because a screening modality's clinical utility is primarily based on how well it can identify tiny, early-stage tumours—precisely when intervention is most effective and survival outcomes are most favorable. The ethical imperative is clear: a screening tool that performs well on advanced cancers but fails to detect early-stage disease would provide limited clinical benefit and could potentially harm patients through false reassurance. It is unclear without this stratified analysis whether the model's performance is largely driven by the detection of larger, more advanced cancers or is consistent across different stages. In order to more accurately evaluate the model's potential for early detection, it is imperative that future research validate the model using a clinical dataset that contains this specific information.

4.2.5 Hardware and Resource Constraints: Although not a methodological weakness, there is a practical constraint related to hardware resources. The training and large-scale validation of deep learning models demand substantial computational power, including access to GPUs or TPUs, which can be cost-prohibitive for many research institutions. Addressing this challenge will be essential for broader validation and eventual clinical translation of such models.

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Data availability

The data that was used to obtain the findings can be found in the publicly available Visual DMR dataset

Declarations

The authors declare that they have no known competing

financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Conflict of interest

The authors do not have any conflict of interest.

Declaration of Generative AI in Scientific Writing

During the preparation of this work, the author(s) used ChatGPT tool to assist with writing process. After utilizing this tool/service, the author(s) reviewed and edited the content as necessary and assume full responsibility for the final content of the published article.

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